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Batch - 3 (Data Science)

CL and NLP
Assignment - 1

Q.1 A scalar LSTM... $x_t = 0.5$, with hidden state $h_{t-1} = 0.2$... Compute $f_t, i_t, \tilde{c}_t, c_t, o_t, i_t$.

A: $x_t = 0.5$, $h_{t-1} = 0.2$, $c_{t-1} = 0.4$

$$f_t = \sigma(W_f \cdot x_t + U_f \cdot h_{t-1} + b_f)$$
$$= \sigma((0.7 \times 0.5) + (0.3 \times 0.2) + 0.1)$$
$$= \sigma(0.5100)$$

$$\therefore \sigma(0.5100) = \frac{1}{1 + e^{(-0.5100)}} = \frac{1}{1.6005} = 0.6248$$

$$\therefore f_t = 0.6248$$

$$i_t = \sigma(W_i \cdot x_t + U_i \cdot h_{t-1} + b_i)$$
$$= \sigma((0.6 \times 0.5) + (0.4 \times 0.2) + 0.0)$$
$$= \sigma(0.3800)$$

$$\therefore \sigma(0.3800) = \frac{1}{1 + e^{-(0.3800)}} = \frac{1}{1.6839} = 0.5939$$

$$\therefore i_t = 0.5939$$

$$\tilde{c}_t = \tanh(W_c \cdot x_t + U_c \cdot h_{t-1} + b_c)$$
$$= \tanh((0.9 \times 0.5) + (0.2 \times 0.2) + 0.0)$$
$$= \tanh(0.4900)$$

$$\therefore \tanh(0.4900) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$
$$= \frac{e^{(0.4900)} - e^{-(0.4900)}}{e^{(0.4900)} + e^{-(0.4900)}} = 0.4542$$

$$\therefore \tilde{c}_t = 0.4542$$

$$O_t = \sigma(W_o \cdot x_t + U_o \cdot h_{t-1} + b_o)$$
$$= \sigma((0.5 \times 0.5) + (0.5 \times 0.2) + 0.1)$$
$$= \sigma(0.4500)$$

$$\therefore O_t(0.4500) = \frac{1}{1+e^{-(0.4500)}} = 0.6106$$

$$\therefore O_t = 0.6106$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$
$$= (0.6248 \times 0.4) + (0.5939 \times 0.4542)$$
$$= 0.5197$$

$$h_t = \tanh(c_t) \odot O_t$$
$$= \tanh(0.5197) \times 0.6106$$
$$= 0.4775 \times 0.6106$$
$$= 0.2916$$

Q2 a) How many trainable parameters does the layer have?

A. I/P Dimension ($x_{dim.}$) = 10
Hidden Units ($h_{dim.}$) = 32

For a single gate, linear transformation is :

$$gate = \sigma(Wx_t + Uh_{t-1} + b)$$

I/P Weight : Parameters = $h_{dim.} \times x_{dim.}$
$= 32 \times 10 = 320$

Recurrent / Hidden Weight : Parameters = $h_{dim} \times h_{dim.}$
$= 32 \times 32 = 1024$

Bias : One bias per hidden unit
Parameters = hdim. = 32

Total Trainable Parameters = 320 + 1024 + 32 = 1376

∴ We have 4 gates.
∴ Total Trainable Parameters = 4 × 1376
= 5504

b) How many trainable parameters does the layer have under that convention?

A: In such frameworks, the transformation for each gate includes two independent bias — one for input projection and one for hidden state projection.

I/P Weight = 32 × 10 = 320
Hidden Weight = 32 × 32 = 1024
I/P Bias = 32
Hidden Bias = 32

Total Parameters per gate = 320 + 1024 + 32 + 32 = 1408

∴ Total Trainable Parameters = 4 × 1408
= 5632

Q3a) for k = 10 steps
Scaling Factor = 0.9^{10}
≈ 0.3487

For $k = 50$ steps

Scaling factor $= 0.9^{50}$

≈ 0.0052

b) For $f = 0.5$

Scaling Factor $= 0.5^{50}$

$\simeq 0.8882 \times 10^{-15}$

For $f = 0.99$

Scaling Factor $= 0.99^{50}$

$\simeq 0.6050$

c)
- Smaller forget gate values causes the gradient to vanish quickly over 50 steps, which makes learning long-term dependencies impossible.
- LSTM solves this by learning to keep the forget gate value close to 1.0. when memory needs to be preserved, which creates a direct linear highway for the gradient to flow backward through the cell state.
- Vanilla RNN lack this path and multiply gradients through repeating non linear functions, causing them to vanish completely

Q4a) Total Hourly Records = Years x Day/Year x Hour/Day

$= 3 \times 365 \times 24$

$= 26280$

b) Total I/P / O/P Window →

Total Window Size = 168 + 24 = 192 hrs

(168 ↓ 100kblocks, 24 ↓ prediction)

⇒ Total Records = Total Window Size + 1

⇒ 26280 − 192 + 1

⇒ 26089

c) With Batch Size = 64

~~Shape format~~ = (Batch Size, Time Step, Features)

I/P Batch Shape = (64, 168, 3)

O/P Batch Shape = (64, 24, 1) – If predicting only 1 feature

If predicting every feature

= (64, 24, 3).

d) Training Records = 70% × 26280

$= \frac{70}{100} \times 26280 = 18396$ records

Training Windows = 18396 − 192 + 1

= 18205 windows

Mini Batches per Epoch = $\frac{18205}{64}$

= 284.45

≃ 285 (including Partial Batch).

Parameter for 1 Gate :
Input Weights : $64 \times 3 = 192$
Recurrent Weight : $64 \times 64 = 4096$
Bias : 64

Total Per Gate : $192 + 4096 + 64$
$= 4352.$

∵ 4 Gates are there,
∴ Total trainable parameters $= 4 \times 4352$
$= 17408$ Parameters.

25 a) Min-Max Normalization $= \frac{x - x_{min}}{x_{max} - x_{min}}$

Normalized Value $= \frac{95 - 20}{120 - 20}$

$= \frac{75}{100} = 0.75$

b) The O/P Probability $= 0.83$
Threshold $= 0.5$
∵ O/P > Threshold, so model predicts a failure (Negative class).

c) True Positive (TP) = 30 (correctly flagged failure)
False Positive (FP) = 45 - 30 = 15 (falsely flagged failure)
False Negative (FN) = 40 - 30 = 10 (missed actual failure)
True Negative (TN) = 1000 - 30 - 15 - 10 = 945 (correctly flagged normal)

Precision $= \frac{30}{45} = 0.667 \simeq 66.7\%$.

Recall $= \frac{30}{40} = 0.750 \simeq 75\%$.

Accuracy $= \frac{30+945}{1000} = \frac{975}{1000} = 0.975 \simeq 97.5\%$.

F1 Score $= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

$= 2 \times \frac{0.667 \times 0.750}{0.667 + 0.750} = 0.7061 \simeq 70.61\%$.

d) The team should prioritize recall, missing a real pump failure causes costly downtime/ catastrophic damage.

Accuracy is misleading here due to severe class imbalance, only 4% of the total dataset contain actual failures. A broken model predicting "Not Failure" every time still gets a High 96% accuracy.